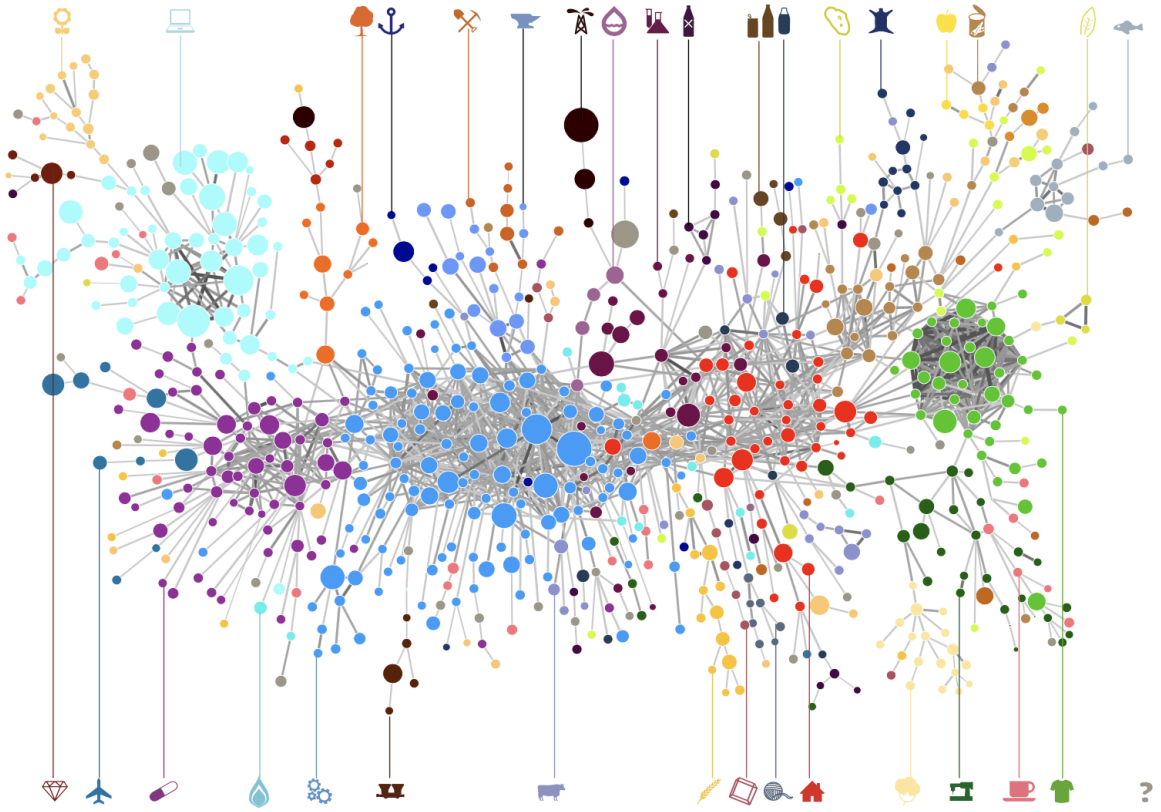


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Optimal Industry Development Strategy Using Economic and Product Complexity

Morocco

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The Economic Complexity Lab - Columbia University

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Optimal Industry Development Strategy for Morocco

0. Executive Summary

In this report, we aim to uncover the economic path Morocco went through during the period from 2019 to 2023. Specifically, we dive into the topic through a macroeconomic overview of Morocco from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We find that during the period of our interest, Morocco demonstrated a moderate level of economic growth with an average annual GDP growth rate of 2.63% between 2019 and 2023, with a similar economic complexity compared to other emerging economies.

The mixed progress during the period, we believe, can be partially explained through the analysis of the economic and product complexity structure of Morocco on the industry group level, as well as the declining macro and microeconomic strategies. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that Morocco progressed in developing most of its industries during the period while its emerging counterparts and economies around the globe also adopted complex and progressive paths with variable degrees of success.

Based on our analysis of the opportunity gain and sensitivity analysis of major industry groups, Morocco could have furthered its gains by prioritizing Energy Drilling & Mining, Services & Utilities, and Precious Stones. We further note that Morocco has a slightly mismatched development strategy that indicates a moderate level (compared to other emerging countries) of structural optimality on the complexity level, having an adjusted R-square of around -0.9%. Since a significant positive correlation exists between GDP growth and the structural optimality index of an economy as we define and calculate, Morocco indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

1. Report Overview

In this report, we aim to uncover the economic path Morocco went through during the period from 2019 to 2023. Specifically, we dive into the topic through a macroeconomic overview of Morocco from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We then analyze the industrial framework and changes of Morocco during the period based on quantitative measures such as the Product Complexity Index (PCI) and Product Family Complexity Index (PfCI), followed by a detailed analysis of how such changes have matched with development strategies as quantitatively measured by opportunity gains on an industry-level. We conclude our report with macro-level results and predictions of Morocco, as deduced from regression models we have established on a global scale, which take into account countries' stances in their product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

Figure 1a presents a time-series comparison of GDP (in USD billions) for Morocco and three peer countries—South Africa, Egypt, and Tunisia—from 2019 to 2023. Over this five-year period, Morocco exhibited a relatively steady upward trend in GDP, contrasting with the more volatile trajectories of South Africa and Egypt. While South Africa's GDP experienced notable fluctuations and Egypt's GDP peaked in 2022 before declining in 2023, Morocco's growth remained more consistent and gradual. Tunisia's GDP remained significantly lower than its peers, with only slight increases. Morocco's average annual GDP growth rate was 2.63%, placing it above South Africa (0.32%) but below Egypt (5.26%). This reflects a more stable and resilient growth pattern, especially when compared to South Africa's stagnation and Egypt's sharp rise and subsequent decline. Morocco's performance also closely mirrors that of Tunisia (3.03%), though at a higher absolute GDP level.

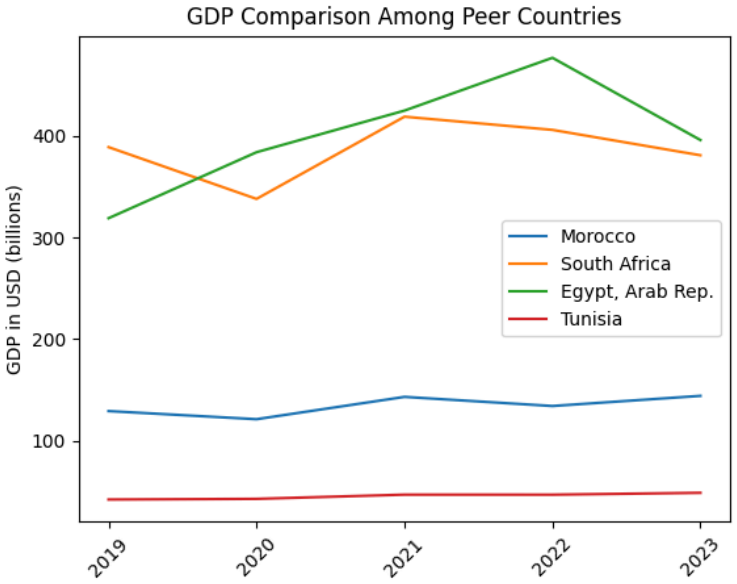


Figure 1a. GDP Comparison Among Peer Countries (2019-2023)

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), developed by Ricardo Hausmann and Cesar Hidalgo in 2009, measures the amount of productive knowledge embedded in a country's economy. As illustrated in Figure 2a, Morocco experienced a notable decline in production capability from 2019 to 2023, with its ECI falling from +0.61 in 2019 to -0.70 in 2023—a drop of 1.31 index points. This indicates a significant loss in economic complexity, suggesting a shift toward less diverse or less sophisticated exports over the five-year period.

In comparison, South Africa and Egypt both maintained stronger positions. South Africa saw a recovery and overall improvement in ECI after 2019, peaking in 2021 and stabilizing through 2023. Egypt exhibited modest but positive growth, with its ECI improving steadily, albeit at a slower rate. Tunisia, like Morocco, experienced a decline, though less severe. Morocco's performance places it in a relatively weak position among its peers in terms of economic complexity by 2023.

While the comparable countries exhibited fluctuations between 2019 and 2020, the most prominent shifts occurred between 2020 and 2021—likely reflecting the uneven economic responses to the COVID-19 pandemic. Morocco's continued decline beyond 2021 may reflect structural vulnerabilities or reduced investment in export diversification and innovation. Overall, the trajectory highlights Morocco's growing challenge in sustaining productive knowledge and technological capacity relative to regional peers.

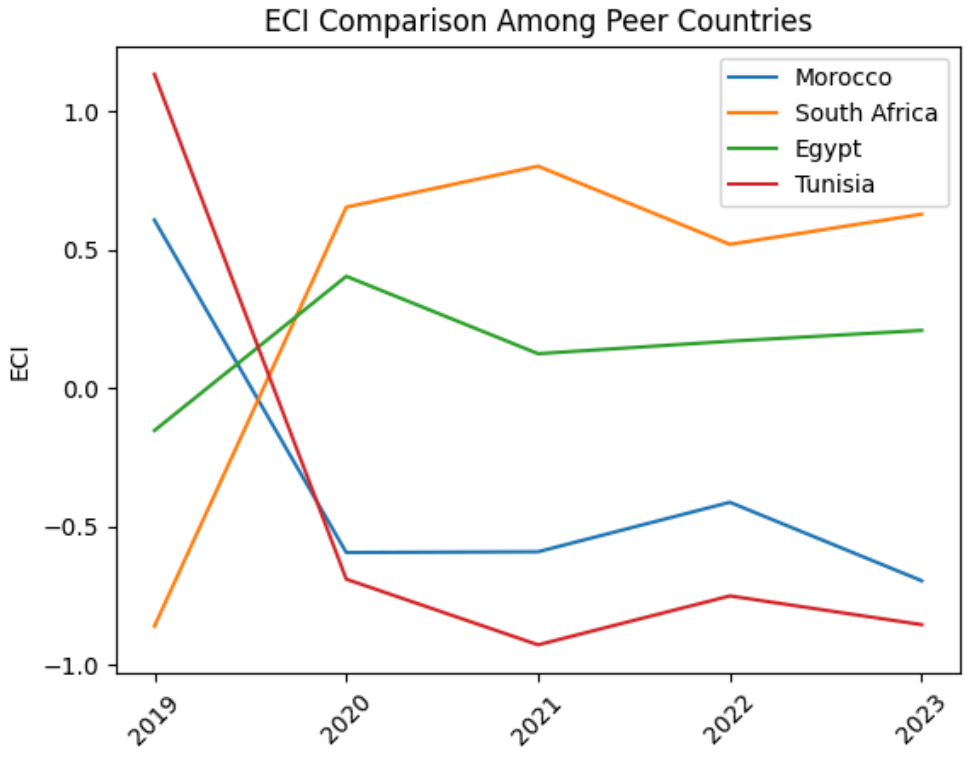


Figure 2a. ECI Comparison Among Peer Countries (2019-2023)

Figure 3a presents a joint analysis of GDP and the Economic Complexity Index (ECI) for Morocco between 2019 and 2023. While Morocco's GDP exhibited a volatile yet generally upward trend—recovering strongly after a drop in 2020—its ECI followed a contrasting path. The ECI declined sharply from +0.61 in 2019 to -0.70 in 2023, reflecting a steady loss in economic complexity even as GDP rebounded.

This divergence suggests that GDP growth during this period was not consistently aligned with increases in productive knowledge or export sophistication. In particular, from 2019 to 2021, both GDP and ECI declined, possibly indicating broader economic contractions affecting both output and complexity. However, while GDP recovered in 2021 and continued to rise, ECI failed to rebound and instead continued its downward trajectory.

These results highlight a disconnect between Morocco's short-term economic growth and long-term productive capacity. While GDP gains might stem from cyclical factors or external demand (e.g., resource exports or tourism), the persistent decline in ECI points to deeper structural challenges in upgrading the country's production and export base. In the following sections, we explore Morocco's production structure and product space to better understand the sources of this divergence and evaluate opportunities for improving its economic complexity.

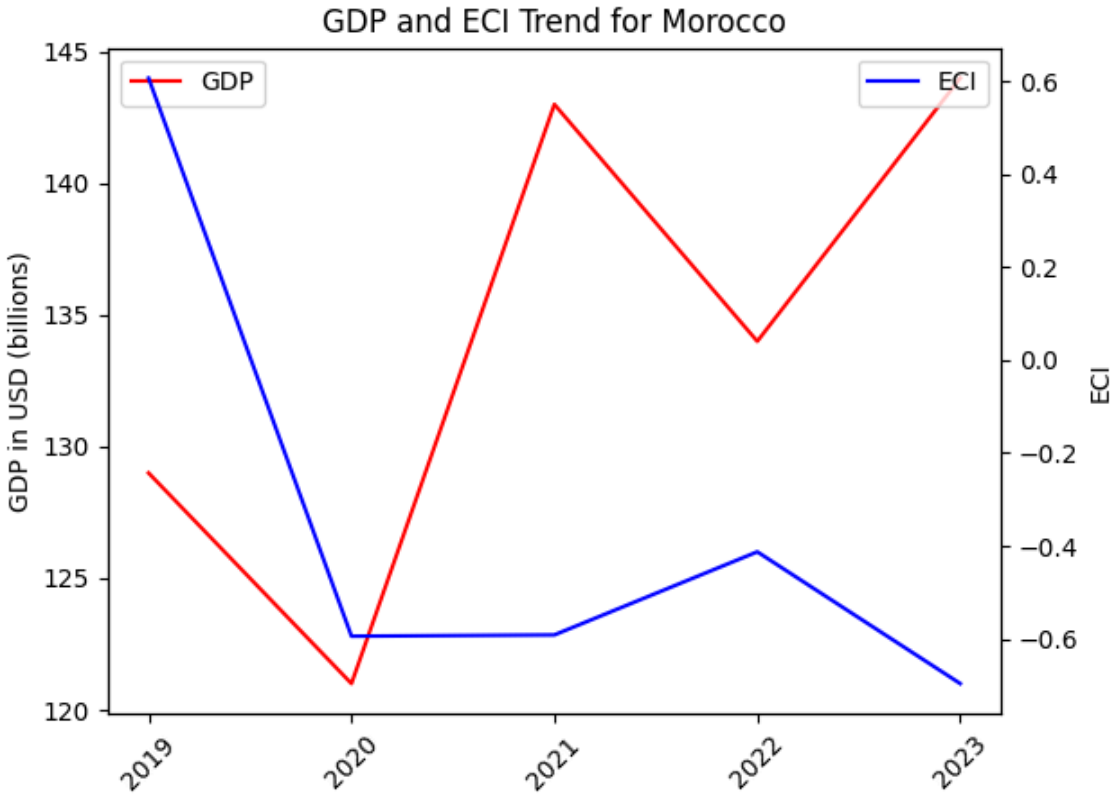


Figure 3a. GDP Growth and ECI for Morocco (2019-2023)

3. Production Analysis

3.1 Product Categorization

Our main data source in this research is the **UN Comtrade Website**, which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Products in our data source are labeled from 010000-999999, with the first two digits representing a general category to which the product belongs. Based on the 99 categories extracted from the label, we further group the products into the 18 most widely-accepted industries, as summarized in Table 1.

Note that the High-complexity Products industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technological gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), the High-complexity Products industry has the highest PfCI among all industry groups.

Category	Description
Animals and By-products	All kinds of meat, livestock and related by-products
Vegetables, Fruits and Plants	All kinds of plants, edible vegetables and fruits
Food and Preparations	Edible preparations, raw cooking materials and appliances
Clothing and Accessories	All kinds of fabrics, filament, clothing and accessories
Textiles	All kinds of textile articles and related products
Wood Products	Book, paper and other elementary products related to wood
Metals	Mining of all sorts of raw Metals
Refined and Processed Stones	Refined or processed stone products such as pearl and ceramics
Energy Drilling and Mining	Mining industry for energy, minerals, stone materials, etc.
Daily Instruments	Basic daily instruments such as sport requisites, timing and measuring tools
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Chemicals	Raw chemical materials and by-products
Machinery and Construction	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapons	Explosives, arms and ammunition
Services and Utilities	Financial, business, legal, communication, travel Services and Utilities, etc.
High-complexity Products	Agglomeration of advanced products with highest PCIs

Table 1. Categorization of Products

We further observe a positive correlation between the GDP and the export of High-complexity Products. A higher adjusted R-square exists between GDP and High-complexity Products exports, indicating that more successful economics, as measured by GDP, tend to be those that are capable of producing and exporting High-complexity Products as we define them. Tables 2a and 2b show the OLS results in a universe of 107 and 21 typical countries respectively.

	High-complexity Products	Food and Preparations	Animals and By-products
Adj. R-squared	0.075	0.006	0.002
F-statistic	9.556	1.638	1.175
No. Observations	107	107	107

Table 2a. OLS Regression Results of GDP against Percentage Volume of Exports

	High-complexity Products	Food and Preparations	Animals and By-products
Adj. R-squared	0.149	0.003	-0.012
F-statistic	4.510	1.059	0.7542
No. Observations	21	21	21

Table 2b. OLS Regression Results of GDP against Volume of Percentage Exports

3.2 Realized Industry Development

Based on the aforementioned product categorization, we aim to assess Morocco's realized industry production strategy between 2019 and 2023 through the lens of economic complexity. In Hausmann and Hidalgo (2009), *distance* is defined as a measure of how far a country is from being able to competitively produce a given product. We extend this concept to broader product groups to quantify how technically prepared Morocco is to develop each category. A shorter distance reflects greater capability, and we define the net change in distance over time as Realized Industry Development (RID). A positive RID suggests progress, while a negative RID indicates regression.

Between 2019 and 2023, Morocco experienced negative RID values across all 18 product groups, as shown in Table 3, indicating a generalized weakening of industrial proximity. The most pronounced declines occurred in precious stones (-0.136), animals and byproducts (-0.120), clothing and accessories (-0.112), and energy drilling and mining (-0.111). Other sectors such as machinery, metals, chemicals, and transportation also saw notable setbacks. This pattern suggests Morocco has become less equipped to competitively develop across its industrial landscape during this period.

Category	RID
Precious Stones	-0.136093
Animals and By-products	-0.119704
Clothing and Accessories	-0.112472
Energy Drilling and Mining	-0.111039
Metals	-0.104850
Vegetables, Fruits and Plants	-0.101941
Daily Instruments	-0.101414
Machinery and Construction	-0.096526
Raw Manufacturing	-0.095086
Chemical Manufacturing	-0.094941
Services and Utilities	-0.093107
Weapons	-0.091381
High-complexity Products	-0.088247
Chemicals	-0.071089
Wood Products	-0.053968
Textile	-0.052644
Transportation	-0.051892
Food and Preparations	-0.021005

Table 3. Realized Industry Development (RID) of Morocco

3.3 Regional and Global Comparison

Table 4a compares Morocco's production strategy with that of leading economies—France, Greece, and Portugal—using Realized Industry Development (RID) across sectors. France showed the strongest focus on complex industries, with high RID in services and utilities (0.254), chemical manufacturing (0.206), and transportation (0.189). Greece followed similar trends, though with lower RID values, especially in animals and by-products (0.089) and vegetables, fruits, and plants (0.091). Portugal, by contrast, saw negative RID in nearly all sectors, notably in machinery and construction (-0.163) and transportation (-0.175), indicating a retreat from advanced production. These differences highlight Morocco's relative underperformance in complexity-driven sectors compared to its European peers.

	Category	France, Monaco RID	Greece RID	Portugal RID
8	Clothing and Accessories	0.07685250636	0.04208352226	-0.07281306464
13	Weapons	0.0870084023	0.0825859352	-0.05860993721
10	Metals	0.1002541169	0.03826552211	-0.06089258102
4	Chemicals	0.1089481024	0.04345801555	-0.06914017018
16	High-complexity Products	0.109929078	0.01068895643	-0.1531408308
0	Animals and By-products	0.1127753899	0.08944437274	-0.05061572677
11	Machinery and Construction	0.1456790757	0.03421908844	-0.1625880239
1	Vegetables, Fruits and Plants	0.1521817777	0.09113531081	-0.0434734408
7	Wood Products	0.1550820874	-0.01879599805	-0.0554056145
2	Food and Preparations	0.1587268475	0.04716530365	-0.09267542032
15	Precious Stones	0.1673832841	0.05994466173	-0.05314976252
9	Textile	0.1756877154	0.04521006518	-0.04347842216
14	Daily Instruments	0.1786019317	0.01459782316	-0.09267394417
3	Energy Drilling and Mining	0.1872416606	0.03492697038	-0.1492010129
12	Transportation	0.1890617755	0.05140105927	-0.1746151433
6	Raw Manufacturing	0.1895080092	0.02492461073	-0.1459429103
5	Chemical Manufacturing	0.206177492	0.03944065704	-0.1526027583
17	Services and Utilities	0.2543037567	0.04875418738	-0.083337972

Table 4a. RID Comparison across Leading Economies (2023)

Table 4b further evaluates Morocco's development strategy in comparison to emerging global economies, specifically China and Singapore. These countries have focused their efforts on intermediate and high-complexity industries, with China showing moderate positive RID across most sectors—particularly in services and utilities (0.101), machinery and construction (0.070), and chemicals (0.056). This reflects a steady move toward higher industrial complexity. In contrast, Singapore's RID values are more mixed, with notable gains in high-complexity products (0.076) but declines in key sectors such as vegetables, fruits, and plants (-0.053) and transportation (-0.025). These patterns suggest that while China is gradually strengthening its industrial capabilities across a broad range of sectors, Singapore's approach has involved more selective shifts, emphasizing advanced products but retreating in others. Morocco's own RID trajectory contrasts with these emerging peers, signaling fewer gains and a need for more targeted complexity-driven development.

	Category	China RID	Singapore RID
12	Transportation	-0.0248044405	0.05034631924
7	Wood Products	-0.003022440934	-0.02214835676
6	Raw Manufacturing	0.002226687319	0.03758357608
9	Textile	0.02119265495	-0.04881417478
5	Chemical Manufacturing	0.02184597019	0.01889638944
1	Vegetables, Fruits and Plants	0.02280471782	-0.0534278807
10	Metals	0.02494717403	-0.006649662016
3	Energy Drilling and Mining	0.02573971579	-0.05111837385
14	Daily Instruments	0.03454968544	-0.04182397602

8	Clothing and Accessories	0.03510871338	-0.03662171949
15	Precious Stones	0.03695685713	-0.02354349679
2	Food and Preparations	0.03740012745	-0.0404246657
16	High-complexity Products	0.04599797366	0.07634245187
0	Animals and By-products	0.04893135335	-0.0454293208
4	Chemicals	0.05577555955	-0.02762988636
13	Weapons	0.05848765331	-0.008243694737
11	Machinery and Construction	0.07025536931	0.02400148502
17	Services and Utilities	0.1005709689	-0.04195860929

Table 4b. RID Comparison across Emerging Economies (2023)

4. Product Space Design

4.1 Suggested Development Vector

As outlined above, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined in Hausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP = 0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter of the product as compared to the global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in terms of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP = 1) has an opportunity gain of zero. An unreasonable investment in a certain product p would lead to a relative decrease in the export of more profitable products and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economic strength, those having higher ECI, or in other words, more reasonable economic structures are often in better shape.

	Morocco	South Africa	Egypt	Tunisia
0	Energy Drilling and Mining	High-complexity Products	Animals and By-products	Services and Utilities
1	Services and Utilities	Services and Utilities	Wood Products	Textile
2	Precious Stones	Animals and By-products	Weapons	Chemical Manufacturing
3	Machinery and Construction	Transportation	Machinery and Construction	Weapons
4	Weapons	Chemical Manufacturing	Services and Utilities	High-complexity Products
5	Textile	Daily Instruments	Transportation	Precious Stones
6	Wood Products	Chemicals	Daily Instruments	Animals and By-products
7	High-complexity Products	Textile	High-complexity Products	Machinery and Construction
8	Daily Instruments	Machinery and Construction	Vegetables, Fruits and Plants	Energy Drilling and Mining
9	Raw Manufacturing	Clothing and Accessories	Food and Preparations	Wood Products
10	Metals	Raw Manufacturing	Energy Drilling and Mining	Vegetables, Fruits and Plants
11	Animals and By-products	Vegetables, Fruits and Plants	Chemicals	Food and Preparations
12	Vegetables, Fruits and Plants	Food and Preparations	Chemical Manufacturing	Chemicals
13	Food and Preparations	Energy Drilling and Mining	Raw Manufacturing	Raw Manufacturing
14	Chemicals	Wood Products	Clothing and Accessories	Clothing and Accessories
15	Chemical Manufacturing	Metals	Textile	Transportation
16	Clothing and Accessories	Weapons	Metals	Daily Instruments
17	Transportation	Precious Stones	Precious Stones	Metals

Table 5. Opportunity Gain Ranking on a Comparable Scale (2023)

We define the ranking of opportunity gain for a country cc as the Suggested Development Vector (SDV) of cc. Table 5 summarizes how the SDV of Morocco compares with a selected group of regional peers—South Africa, Egypt, and Tunisia—with industry groups ranked in decreasing order of significance. In this analysis, dark blue indicates positive opportunity gains, light blue represents zero gain, and white reflects negative gains. The results show that these comparable countries still identify a significant number of industries with positive opportunity gain, signaling that their economies remain in active transition phases. Among them, Morocco exhibits a development profile similar to its peers, with common areas of opportunity centered on industries such as machinery and construction and chemical manufacturing. This suggests that Morocco continues to share the characteristics of developing regional economies, with a moderately balanced structure that leans toward intermediate-complexity industries and reflects ongoing efforts to expand productive capabilities.

4.2 Budgeted Industry Development Strategy and Interpretation

The optimal budgeted industry development strategy aims to solve the most efficient allocations under the budgeted constraint of different situations of heuristic total export growth. In order to obtain such an allocation portfolio, we formulate an optimization problem as the following:

$$\max_{x \in \mathbf{R}^{|P|}} \quad \left(\frac{k(x) - < k(x) >}{stdev(k(x))}\right)_C$$

$$\text{Subject to}$$

$$\sum_{p \in P} x_p \leq B$$

$$x \succcurlyeq 0$$

where k(x) is the second largest eigenvector of $M_{CC'}$ after adding x to the original export vector, C is the country that we are interested in, P is the product space, B is the heuristic budget.

If we can solve this problem, then we can know what the largest ECI to which a country can improve. Accordingly, the optimal solution x * that we obtain here is exactly our optimal allocation. However, the biggest problem is that the objective function is not convex. Therefore, we cannot apply the classical tools for convex optimization to this problem. In order to solve this problem, our idea is to use a greedy algorithm. The way that we apply the greedy algorithm is that firstly we separate the total budget into k equal parts, and then for the first equal part we try to search for the product that can optimize our objective function by iterating through each $p \in P$. After we find the optimal product p for this equal part, we add the amount back to the export value of this product p and update the ECI for this country. If we observe that ECI does not change after iterating all the products, then we can add up one more equal part to invest. Until we find there is an ECI increment, we add the amount now to that product. The equal rest parts can be done in the same way.

By definition, a greedy algorithm is a kind of local search algorithm that is possible to run into a local minimum. Therefore, for local search algorithms, one typical way to keep it away from the local minimum is to enhance the vision of the algorithm by looking more steps forward. In our setting, whenever we obtain an increment of ECI, we mark down the required amount of export growth. Simultaneously, we continue searching for the next ECI increment point, compare the ratios of these two possibilities and pick the largest one. The ratio is defined as the following:

$$Ratio(p) = \frac{ECI_p - ECI_0}{||x_p - x_0||_1}$$

where ECI_p is just the ECI of the country after adding some amount of export increment to product p, which is exactly the amount that is able to make ECI change. Also, we can express it in the way of $||x_p - x_0||_1$.

In this way, we can obtain a further sighted greedy algorithm. Also, the number of parts that we would like to separate the budget into also matters. In order to solve this problem, we added a stabilizer on top of the enhanced greedy algorithm by running the algorithm for different numbers of parts and selecting the portfolio with the largest ECI increment and most complex economic structure.

Besides, it is quite common that there is a surplus in the budget. To deal with the surplus, we just add it to the product which needs the least amount of dollars to make M_{cp} = 1 (i.e., the smallest amount of required export increase).

Economic Panel

As stated above, the enhanced greedy algorithm is developed mainly for solving the optimal allocation strategy. In order to visualize this allocation, an economic panel is created, consisting of a pie chart, a line chart and a table. First, a pie chart is used to show the percentage of each product in the optimal portfolio. The number located in the center of the circle denotes the heuristic growth of total export, which is also our budget. Each category in the chart has a determined percentage of the allocated budget. The lines below the pie charts show how the ECI changes during the searching process of the greedy algorithm. Even though sometimes an increase of the export by 15% or 20% does not seem reasonable to developed countries, yet it can also give a sense of how a country should develop in the long term. Eventually, the graph in the end is just for numerical explanation for the pie charts.

Interpretation

For the purpose of a clear explanation of the line charts, we label all the products in Table 6. Please refer to this table when trying to understand the investment trajectories in the line charts.

Label	Product Names
1	Animal and By-products
2	Vegetables, Fruits and Plants
3	Food and Preparations
4	Energy Drilling and Mining
5	Chemicals
6	Chemical Manufacturing
7	Raw Manufacturing
8	Wood Products
9	Clothing and Accessories
10	Textile
11	Metals
12	Machinery and Construction
13	Transportation
14	Weapons
15	Daily Instruments
16	Precious Stones
17	High-complexity Products
18	Services and Utilities

Table 6. Products Labeling

Below is the economic panel for Morocco in 2023. The four line graphs demonstrate how varying levels of heuristic total export growth—5%, 10%, 15%, and 20%—impact Morocco's Economic Complexity Index (ECI). Across all scenarios, Morocco shows a notable increase in ECI, suggesting strong potential for further economic development. Most importantly, the data indicates that higher export growth rates lead to greater improvements in ECI, implying that Morocco has considerable room for enhancement in both the short and long term.

- At 5% growth, the increase in ECI is largely driven by Precious Stones.
- At 10% growth, gains are attributed to Food and Preparation.
- At 15% growth, the most significant contributions come from Energy Drilling and Mining
- At 20% growth, the ECI gains stem from Energy Drilling and Mining and Precious Stones

These trends underscore the importance of strategically investing in high-opportunity industries to accelerate Morocco's economic complexity and global competitiveness.



Figure 4. Greedy-Based ECI Improvement Results of Morocco with Different Budgets (2023)

Morocco (2023)				
Original ECI	0.607076			
Heuristic Export Growth	**5%**	**10%**	**15%**	**20%**
Food and Preparation (USD)	682,056,870.1	3,239,770,133	323,977,013.3	1,341,378,511
Energy Drilling and Mining (USD)	0	0	6,155,563,252	8,718,960,322
Textile (USD)	341,028,435	341,028,435	647,954,026.6	1,341,378,511
Precious Stones (USD)	1,534,627,958	1,534,627,958	2,591,816,106	8,718,960,322
Increased ECI	-0.07412158461	-0.07412158461	0.1938427259	0.4809992351

Table 7. Portfolio of Morocco (2023)

4.3 Sensitivity Analysis and Contingency Strategy

In the following section, our primary objective is to assess Morocco's sensitivity to changes in export performance across various industries. While the Economic Complexity Index (ECI) provides valuable insights into a country's productive capabilities, less attention has been given to the stability of the index itself. This analysis aims to identify the key drivers of ECI stability and offer a deeper understanding of long-term ECI growth trajectories. Furthermore, visualizing industry-specific sensitivity helps develop intuition about which

sectors are currently supporting economic resilience and which offer the greatest returns on investment, thereby guiding strategic development priorities.

ECI Sensitivity Graph

First, in order to assess the effect particular industries have on each respective country, we developed ECI Sensitivity graphs (Figure 5). We use the shocking methodology to increment or decrement an industry's export level by a certain percentage and recalculate the ECI, all else being equal. While freezing the export dynamics of the rest of the world and enacting an instantaneous shock in export for a particular industry and country may not reflect real-life dynamics, what it does offer us is an understanding of a country's level of dependency on an industry in the context of ECI. The shocking is done in small percentage increments in order to smooth out the resulting curve. Additionally, a recurring theme one will see is that for most industries, a breaking point occurs where the ECI either jumps upwards or downwards, creating a hockey stick shape rather than a linear trend. While the inflection point may not be accurate in the context of real life, comparing the various inflection points across industries can give us valuable insight.

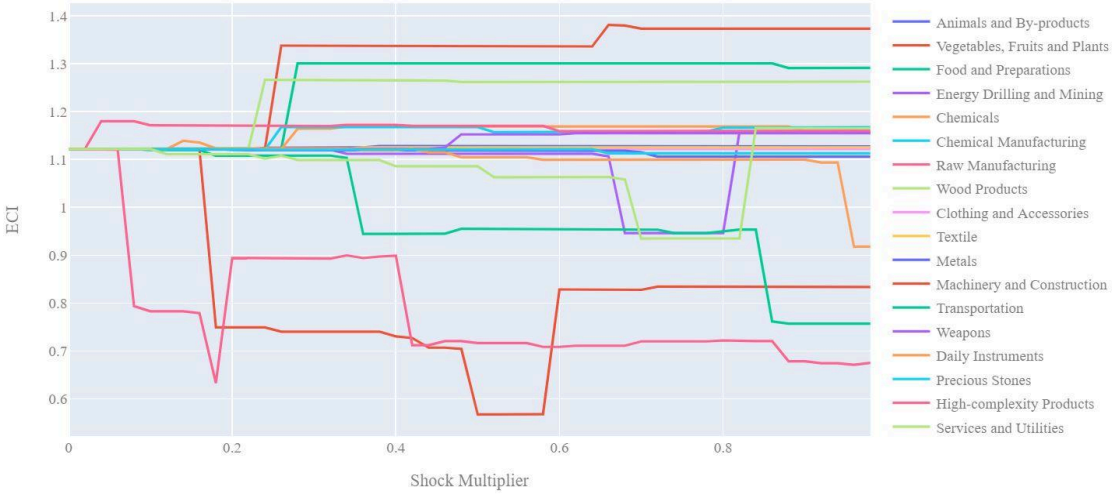


Figure 5. Sample ECI Sensitivity Graph (2022)

The graphs are formulated as follows. There are two ECI Sensitivity graphs. The first chart, Incremental ECI Sensitivity, shows positive shocks in which export numbers are shocked upwards per industry. The X-axis is in terms of percentage shock per current dollar value of export. The Y-axis is the recalculated ECI level post-shocking. The second chart, Decremental ECI Sensitivity, follows the same format except the X-axis represents negative shock. In other words, exports for each industry are decremented, and the ECI is accordingly recomputed.

4.3 Sensitivity Analysis Results and Interpretation

We analyze Morocco's 2023 ECI Incremental and Decremental Sensitivity Charts to pinpoint which sectors wield the greatest influence over its Economic Complexity Index—and what that means for strategic development. On the incremental side (Figure 6a), High-complexity Products deliver the largest ECI change when positively shocked, reflecting their deep linkages to advanced value chains and high-knowledge activities (e.g., advanced manufacturing, ICT-integrated industries). Daily Instruments rank a close second, their strong sensitivity underscoring Morocco's entrenched specialization in precision goods that supply regional manufacturing hubs. Machinery & Construction follows, highlighting growing capabilities in intermediate and capital goods production. Notably, the outsized sensitivity of High-complexity Products aligns with Morocco's recent push into tech-driven sectors, signaling that sustained investment here can unlock disproportionate gains in productive knowledge.



Figure 6a. Incremental ECI Sensitivity for Morocco (2023)

On the decremental side (Figure 6b), Food & Preparations stands out as the most vulnerable: even minor negative shocks to agrifood exports trigger the steepest ECI decline. This reflects structural overreliance on low-complexity agrarian outputs with limited spillover potential. Other sectors—such as textiles, chemicals, and energy drilling—show relatively muted sensitivity, indicating that ECI risks are concentrated in a few critical areas.

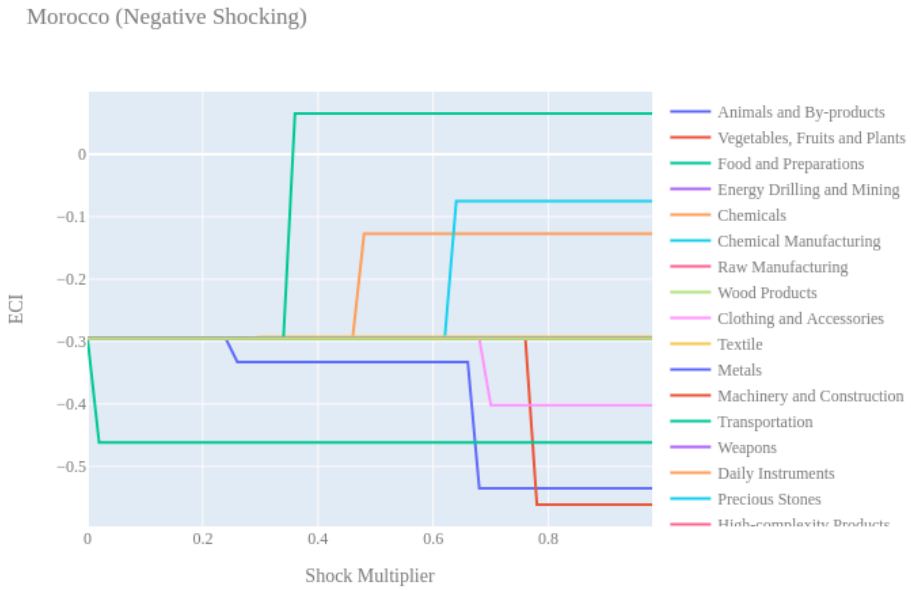


Figure 6b. Decremental ECI Sensitivity for Morocco (2023)

These patterns suggest a dual strategy. First, prioritizing High-complexity Products, Daily Instruments, and Machinery & Construction—the sectors with the highest incremental leverage—can accelerate Morocco's climb up the complexity ladder. Second, reducing agrifood dependency through value-chain upgrading (e.g., food processing innovation, quality certifications) would mitigate downside risks. By targeting high-upside sectors while stabilizing low-complexity exposures, Morocco can solidify its transition toward a more resilient, knowledge-driven economy.

5. Macro-level Predictions

5.1 Structural Optimality Index

ECI sensitivity described in the section before has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our development strategies are and how deviations from the suggested developmental path will lead to suboptimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the Adjusted R-Square obtained from regressing the country's realized industry development over the suggested development strategy. Both strategies are quantified in nature, with the realized development represented by increasing export in every major industry group and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The adjusted R-square from the regression thus provides us with a quantified measure of how the development path adopted by the country has fitted with/differed from calculations on a complexity level.

After calculating the Structural Optimality Index (SOI) for all countries in our database, we regressed five-year GDP growth (2019–2023) against SOI. Figure 7 illustrates a significant positive correlation between countries' SOI and GDP growth, with the regression achieving an adjusted R^2 of -0.9%. This confirms that economies with higher structural alignment to complexity-driven developmental paths tend to achieve stronger growth outcomes.

Table 8: OLS Regression Results	
Dep. Variable: GDP	**F-Statistic:** 0.05747
R^2: 0.001	**Prob (F-Stat):** 0.811
Adj R^2: -0.009	**No. Observations:** 109

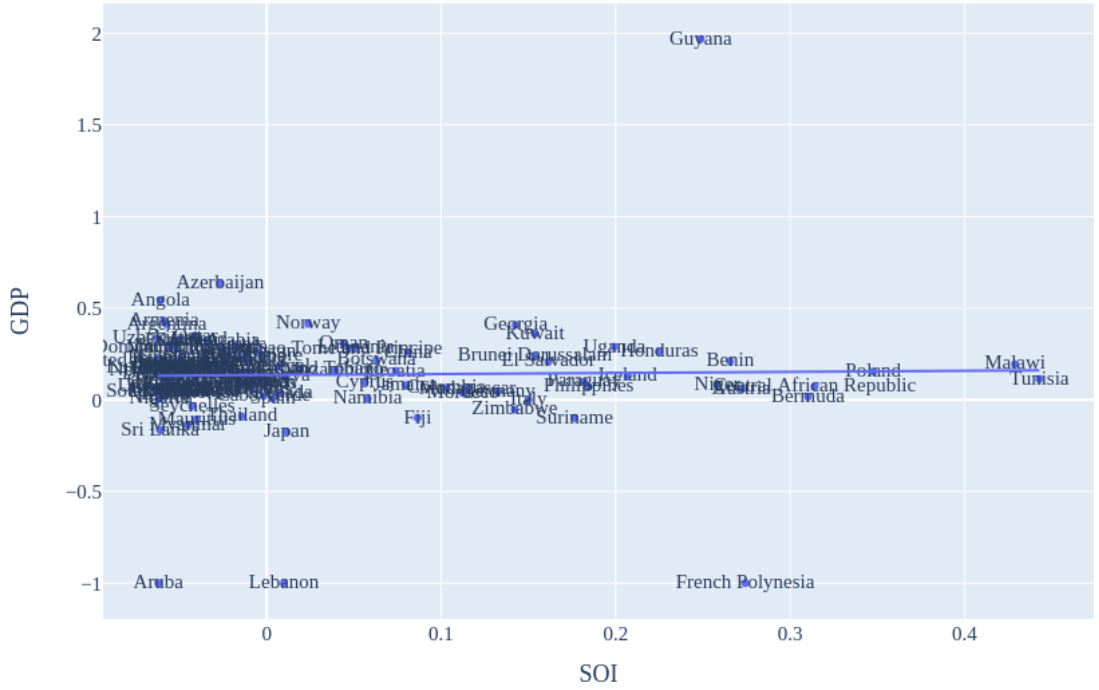


Figure 7. Regression of Five-Year GDP Growth over SOI (2019-2023)

5.2 SOI of Morocco and Its Implications

During the examined period, Morocco exhibits a low SOI compared to developed peers, reflecting structural imbalances. As shown in its SOI decomposition, RID (Revealed Institutional Development) scores are predominantly negative (-0.1 to -0.4), signaling institutional gaps that hinder optimal resource allocation. Opportunity Gain RID metrics are also negative (-0.02 to -0.14), underscoring missed opportunities in sectors like Services and Utilities (-0.01 to -0.14) and Commercials (-0.02 to -0.14). Notably, Food and Preparations (-0.02 to -0.14) remains a critical vulnerability, aligning with earlier ECI sensitivity findings.

Morocco ranks in the third quartile globally for the Structure of Opportunity Index (SOI), with a five-year GDP growth rate of 2.63% (2019–2023), trailing behind high-SOI emerging economies. This relative underperformance stems from a development strategy misaligned with complexity-driven priorities. For instance, Morocco has overinvested in sectors like Food and Preparations—a low-opportunity-gain category with a negative Realized Industry Development (RID) of -0.0210—while under prioritizing more promising sectors such as Services and Utilities (RID: -0.0931, but high opportunity gain) and High-Complexity Products (RID: -0.0882), which exhibit stronger upward potential despite recent underperformance.

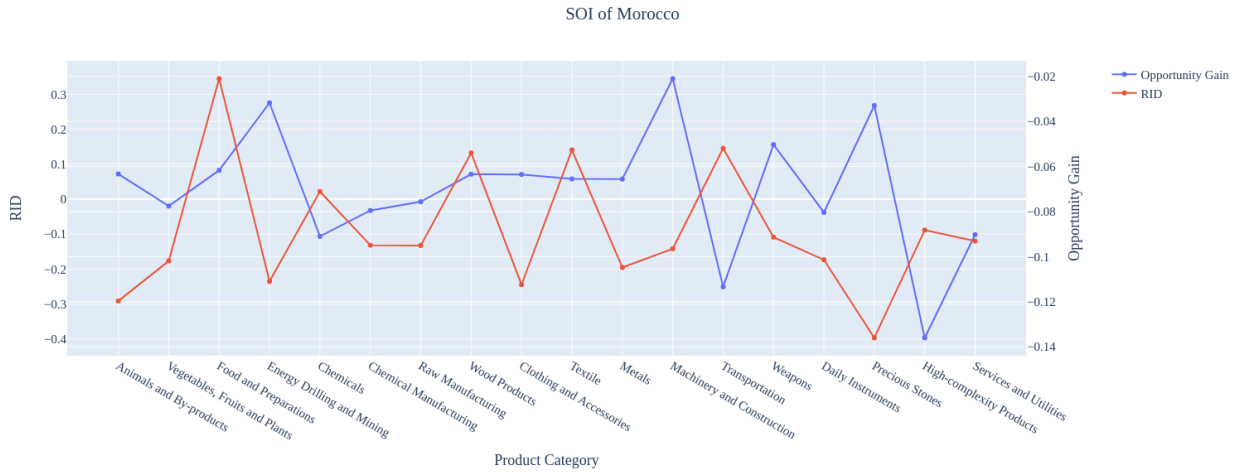


Figure 8: SOI for Morocco (2019-2023)

Between 2019 and 2023, Morocco maintained a strategic focus on Food and Preparations and Textiles (RID: -0.0526), both of which now appear overdeveloped relative to their opportunity profiles. In contrast, sectors such as Services and Utilities, Commercials, and High-Complexity Products have shown substantial opportunity gains in Morocco's Suggested Development Vector (SDV), indicating untapped potential for knowledge accumulation and economic upgrading.

6. Concluding Remarks

In conclusion, Morocco experienced moderate economic growth from 2019 to 2023, with a five-year GDP growth rate of 2.63% and a stagnating trajectory in economic complexity, especially when compared to faster-growing economies in its regional and global peer group. While Morocco's Economic Complexity Index (ECI) showed limited upward movement, our analysis suggests that much of this underperformance is attributable to misaligned industry-level development priorities.

Using our industry group classification system, we measured the Realized Industry Development (RID) across sectors and observed that Morocco disproportionately focused on Food and Preparations, Clothing and Accessories, and Textiles—all of which exhibited low opportunity gains and negative RID scores during this period. In contrast, peer economies and global leaders increasingly prioritized more complex industries, including Services and Utilities and High-complexity Products, aligning their development strategies with sectors offering greater structural transformation potential.

Drawing from our Suggested Development Vector (SDV) and industry sensitivity analyses, we find that Morocco stands to gain by reallocating investment toward underdeveloped yet high-opportunity sectors such as Services and Utilities, Commercials, and High-complexity Products. These sectors not only exhibit stronger marginal contributions to ECI growth but also align better with global complexity-driven development trends.

We also find that Morocco's adjusted R^2 of approximately -0.009 –representing the correlation between GDP growth and Structural Optimality Index (SOI)– indicates moderate strategic coherence in its industrial policy. However, this remains below the performance of high-SOI economies like Vietnam. Thus, Morocco's recent economic path reflects a moderate level of structural misalignment and a suboptimal development strategy, suggesting considerable room for improvement through targeted industrial diversification and complexity-based planning.

By adjusting its economic structure to better exploit high-return opportunities in complex sectors, Morocco can transition toward more sustainable and inclusive long-term growth, better positioning itself within the global economic landscape.

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